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### Radar sidelobe blanking

#### Abstract

A sidelobe blanking method is used with a radar system with a linear transmit array along a first axis and a linear receive array along a second axis perpendicular to the first axis. The method transmits a transmit guard signal using a transmit guard antenna with a wide beamwidth in both axes, determines a first guard signal strength of a signal captured by the receive array at a first receive angle, transmits a radar signal using the transmit array at a transmit angle, determines a second guard signal strength of a signal captured by a receive guard antenna with a wide beamwidth in both axes, receives a radar return signal using the receive array at the receive angle, and blanks the radar return signal when its signal strength is less than either the first guard signal strength or the second guard signal strength.

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## Background/Summary

### BACKGROUND

(1) Sidelobe blanking is a technique commonly used with some radar systems to discriminate between true return signals and erroneous return signals arising from sidelobe clutter. To implement sidelobe blanking, a guard antenna operates near a primary antenna (e.g., an antenna array or reflector antenna) of a radar system. The guard antenna has a main lobe that is angularly wide enough to cover the main lobe and principal sidelobes of the nearby primary antenna. The guard antenna has a gain that is greater than that of the primary antenna's sidelobes but less than that of the primary antenna's main lobe. When a primary signal outputted by the primary antenna is greater than a guard signal outputted by the guard antenna, it is assumed that the primary signal is a true return signal. However, when the guard signal is greater than the primary signal, it is assumed that the primary signal was produced by sidelobe clutter and is therefore not a true return signal. In this case, the output from the primary antenna may be blanked (e.g., deleted or ignored).

### SUMMARY

(2) In certain embodiments, a sidelobe blanking method is used with a radar system. The radar system has: a plurality of transmit antenna elements forming a transmit array that extends linearly along a first axis; a plurality of receive antenna elements forming a receive array that extends linearly along a second axis perpendicular to the first axis; the transmit array having a first radiation pattern with a main transmit lobe narrow along the first axis and wide along the second axis; and the receive array having a second radiation pattern with a main receive lobe wide in the first axis and narrow in the second axis. The sidelobe blanking method includes transmitting a guard signal from a transmit guard antenna having a wide beamwidth in both the first and second axes; and controlling the receive array to sweep the main receive lobe through a plurality of receive

angles defined relative to the second axis. For each receive angle of the plurality of receive angles, the sidelobe blanking method includes: receiving a first received signal using the receive array; and determining a first guard signal strength of the first received signal for said each receive angle; and rastering the radar system by (i) controlling the transmit array to sweep the main transmit lobe through a plurality of transmit angles defined relative to the first axis and (ii) controlling the receive array to sweep the main receive lobe through the plurality of receive angles. For each transmit angle of the plurality of transmit angles and each receive angle of the plurality of receive angles, the sidelobe blanking method includes: transmitting, with the transmit array, a radar signal; receiving, after said transmitting the radar signal, a second received signal with a receive guard antenna having a wide beamwidth in both the first and second axes; determining a second guard signal strength of the second received signal; receiving, after said transmitting the radar signal, a third received signal with the receive array; determining a main beam signal strength of the third received signal; and blanking the radar signal when the main beam signal strength is less than either the first guard signal strength for said each receive angle or the second guard signal strength.

(3) In certain embodiments, a sidelobe blanking method is used with a radar system. The radar system has: a plurality of transmit antenna elements forming a transmit array that extends linearly along a first axis; a plurality of receive antenna elements forming a receive array that extends linearly along a second axis perpendicular to the first axis. The transmit array has a radiation pattern with a main transmit lobe narrow along the first axis and wide along the second axis; and the receive array has a radiation pattern with a main receive lobe wide in the first axis and narrow in the second axis. The sidelobe blanking method includes: transmitting, with a transmit guard antenna having a wide beamwidth in both axes, a transmit guard signal; receiving a first return guard signal using the receive array with the main receive lobe at one receive angle of a plurality of receive angles defined relative to the second axis; determining a first guard signal strength of the first return guard signal; transmitting a radar signal using the transmit array with the main transmit lobe at one of a plurality of transmit angles defined relative to the first axis; receiving, after said transmitting the radar signal, a second return guard signal with a receive guard antenna having a wide beamwidth in both axes; determining a second guard signal strength of the second return guard signal; receiving, after said transmitting the radar signal, a radar return signal using the receive array with the main receive lobe at the one receive angle; determining a main beam signal strength of the radar return signal; and blanking the radar return signal when the main beam signal strength is less than either the first guard signal strength or the second guard signal strength.

(4) In certain embodiments, a sidelobe blanking method is used with a radar system. The radar system has a plurality of transmit antenna elements forming a transmit array that extends linearly along a first axis and a plurality of receive antenna elements forming a receive array that extends linearly along a second axis perpendicular to the first axis. The transmit array has a radiation pattern with a main transmit lobe narrow along the first axis and wide along the second axis, and the receive array has a radiation pattern with a main receive lobe wide along the first axis and narrow along the second axis. The sidelobe blanking method includes transmitting, with a transmit guard antenna having a wide beamwidth in both axes, a transmit guard signal having first transmit signal characteristics; transmitting a radar signal having second transmit signal characteristics, different from the first transmit signal characteristics, using the transmit array with the main transmit lobe at one of a plurality of transmit angles defined relative to the first axis; receiving a first received signal with the receive array having the main receive lobe at one receive angle of a plurality of receive angles defined relative to the second axis; determining a first guard signal strength of a portion of the first received signal having the first transmit signal characteristics; receiving a second received signal with a receive guard antenna having a wide beamwidth in both axes; determining a second guard signal strength of a portion of the second received signal having the second transmit signal characteristics; extracting, from the first received signal, a radar return signal having the second transmit signal characteristics; and blanking the radar return signal when a

signal strength of the radar return signal is either (a) less than the first guard signal strength, or (b) less than the second guard signal strength.

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## Description

### BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a schematic diagram illustrating one example frequency modulated continuous wave (FMCW) orthogonal linear transmit receive array (OLTRA) radar system with sidelobe blanking, in embodiments.
- (2) FIG. 2 shows the OLTRA antenna of FIG. 1 in further example detail illustrating a first 1D linear array that is orthogonal to a second 1D linear array, in embodiments.
- (3) FIG. 3A shows an example 2D plot of the radar signal of FIG. 1, transmitted by the first 1D linear array, illustrating a main transmit lobe and a plurality of transmit sidelobes, in embodiments.
- (4) FIG. 3B shows an example 2D plot of receive beam characteristics of the second 1D linear array of FIG. 1, illustrating a main receive lobe and a plurality of receive sidelobes, in embodiments.
- (5) FIG. 4 is a 2D plot illustrating a resultant pattern multiplication of the main transmit lobe and the transmit sidelobes of FIG. 3A with the main receive lobe and the receive sidelobes of FIG. 3B, illustrating the spot beam of FIG. 2 and sidelobe products.
- (6) FIGS. 5A and 5B show example perceived targets resulting from processing of radar return signal without sidelobe blanking for a single corner reflector target positioned within a target space of the OLTRA radar system, in embodiments.
- (7) FIG. 6 is a graph illustrating one example fan beam that may represent the transmit main lobe and the receive main lobe of FIG. 2, a transmit guard signal, and a receive guard signal, in embodiments.
- (8) FIG. 7 is a flowchart illustrating one example method for OLTRA radar sidelobe blanking, in embodiments.
- (9) FIG. 8 is a flowchart illustrating one example method for OLTRA radar sidelobe blanking with overlapping transmit signals, in embodiments.
- (10) FIGS. 9A and 9B show example perceived targets resulting from processing of the radar return signal with sidelobe blanking for a single corner reflector target positioned within the target space of the OLTRA radar system, in embodiments.
- (11) FIG. 9B is one example range map plot resulting from processing of radar return signal with sidelobe blanking.
- (12) FIG. 10 is a schematic diagram illustrating an example computing apparatus that implements the signal processor of FIG. 1, in certain embodiments.

### DETAILED DESCRIPTION

- (13) U.S. Pat. No. 8,248,298, incorporated herein by reference for all purposes, discloses an orthogonal linear transmit receive array (OLTRA) radar having two orthogonal linear array antennas that are controlled to scan (e.g., using phase shift) a “pencil” beam over a target area, where the pencil beam is a pattern multiplication of beams of the two orthogonal linear array antennas. A detrimental side effect of the OLTRA radar is erroneous radar return signals caused by sidelobe clutter. Pattern multiplication, previously referred to as cross-product in U.S. Pat. No. 8,248,298, refers to an element-wise multiplication of two antenna beam patterns as used herein.
- (14) FIG. 1 is a schematic diagram illustrating one example of an OLTRA radar system **100** with sidelobe blanking. OLTRA radar system **100** is, for example, a fully integrated radar that includes an OLTRA antenna **104**, a transmitter **130**, a receiver **140**, a signal processor (SIG PROC) **142**, and a radar controller **150**. OLTRA antenna **104** includes a first one-dimensional (1D) linear array **106** (also known as a transmit array) and a second 1D linear array **108** (also known as a receive array)

that is orthogonal to first 1D linear array **106**, where both first 1D linear array **106** and second 1D linear array **108** are independently steerable. OLTRA radar system **100** may operate with a variety of radar waveforms including frequency modulated continuous wave (FMCW), CW and pulse Doppler. Transmitter **130** includes a signal generator **132** that generates a transmit signal that is amplified by an amplifier **131** and fed into first 1D linear array **106**. In the embodiment of FIG. **1**, OLTRA radar system **100** does not require a duplexer because the transmit and receive functions are handled by different apertures (first 1D linear array **106** and second 1D linear array **108**) of OLTRA antenna **104** and switching between transmit/receive modes (as required by a single aperture antenna) is not required. Accordingly, transmitter **130** connects directly to first 1D linear array **106** via a transmit transmission line **162** and receiver **140** connects directly to second 1D linear array **108** via a receive transmission line **163**. Radar controller **150** controls amplifier **131** of transmitter **130** via control line **154** to drive OLTRA antenna **104** and controls OLTRA antenna **104** via control line **156** to output and steer a radar signal **180**. Radar controller **150** also steers second 1D linear array **108** via control line **156** to capture a radar return signal **182**, which may include reflections of the radar signal **180** from an object or target within an operational area of OLTRA radar system **100**, that is fed to receiver **140** via receive transmission line **163**. An output from receiver **140** is fed into signal processor **142** via a transmission line **165**. Signal processor **142** communicates with radar controller **150** via communication channel **166**. An output from radar controller **150** is sent, via a transmission line **168**, to a radar interface **190** and processed by radar software **195** to generate an output on a radar display **192**.

(15) OLTRA radar system **100** also includes a transmit guard antenna **124** that has a wide beamwidth in both axes, is distinct from first 1D linear array **106**, and is driven by a guard transmitter **120** via a transmission line **161** to output a transmit guard signal **184**. Alternatively, transmit guard antenna **124** can be driven by transmitter **130** (e.g., via a switch that connects the output of transmitter **130** to either first 1D linear array **106** or transmit guard antenna **124**). In certain embodiments, transmit guard antenna **124** is implemented by one element of first 1D linear array **106**. In certain other embodiments, transmit guard antenna **124** is implemented by multiple elements of first 1D linear array **106** that are used to generate transmit guard signal **184** as a spoiled beam. Operation of guard transmitter **120** is controlled by a transmit guard control signal **152** received from radar controller **150**. Based on transmit guard control signal **152**, guard transmitter **120** transmits, via transmit guard antenna **124**, transmit guard signal **184**. A first return guard signal **186**, which may include reflections of transmit guard signal **184** from any object or target within the operational area of OLTRA radar system **100**, is received by second 1D linear array **108** and sent to receiver **140** via receive transmission line **163**.

(16) OLTRA radar system **100** also includes a receive guard antenna **174** that has a wide beamwidth in both axes for receiving a second return guard signal **188** and that is coupled, via a transmission line **173**, to a guard receiver **170** that feeds return guard signal **188** to signal processor **142** via a transmission line **175**. In certain embodiments, receive guard antenna **174** is implemented by one element of second 1D linear array **108**.

(17) Signal processor **142** includes a sidelobe blanker **144** that blanks radar return signal **182** when any one or both of first return guard signal **186** and second return guard signal **188** indicates that radar return signal **182** is from a sidelobe and not the main beam, as described in further detail below. In certain embodiments, signal processor **142** is a hard-wired circuit that implements functionality of signal processor **142** described herein. For example, signal processor **142** may be implemented at least in part by an FPGA that is pre-programmed to perform tasks and therefore acts like a hard-wired circuit. Accordingly, in these embodiments the functionality is implemented only in hardware and no machine-readable instructions are included. FIG. **10**, described below, shows an alternative embodiment in which signal processor **142** includes machine-readable instructions. In these embodiments, sidelobe blanker **144** uses two guard signals to determine when the radar return signal would result in clutter on radar display **192**. This overcomes the limitations

of the prior art and provides a complete sidelobe blanking solution for OLTRA radar system **100**. While FIG. **1** shows signal processor **142** as separate from radar controller **150**, it should be understood that signal processor may be integrated with radar controller **150**.

(18) Radar controller **150** coordinates operation of OLTRA radar system **100**, controlling sweeping of OLTRA antenna **104** (both first 1D linear array **106** and second 1D linear array **108**), operation of guard transmitter **120**, synchronization of signal processor **142** to reduce clutter, and output of radar data to radar software **195** for output on radar display **192**. Blanker **144** may also include a thresholding algorithm that determines threshold signal level values of radar return signal **182**, first return guard signal **186**, and second return guard signal **188** for use when determining whether a return in radar return signal **182** indicates a “real” target or whether it results from the sidelobes.

(19) FIG. **2** shows OLTRA antenna **104** of FIG. **1** in further example detail illustrating first 1D linear array **106** orthogonal to second 1D linear array **108**. In the example of FIG. **1**, first 1D linear array **106** is a transmit array formed by sixteen apertures (elements) **202** and second 1D linear array **108** is a receive array formed by sixteen apertures (element) **204**. First 1D linear array **106** transmits radar signal **180**, illustrated as a transmit main lobe **206**, and second 1D linear array **108** receive radar return signal **182**, illustrated as a receive main lobe **208**. Transmit main lobe **206** and receive main lobe **208** represent idealistic main beam characteristics of first 1D linear array **106** and second 1D linear array **108**, respectively. A pattern multiplication of first 1D linear array **106** (transmit main lobe **206**) and second 1D linear array **108** (receive main lobe **208**) is a high-resolution spot beam **210**. In the example of FIGS. **1** and **2**, first linear 1D array **106** has a horizontal orientation (as indicated by horizontal axis **212**) and second 1D linear array **108** has a vertical orientation (as indicated by vertical axis **214**); however, the antenna positions may be switched without change in functionality of OLTRA antenna **104**.

(20) First 1D linear array **106** is electrically driven to produce transmit main lobe **206** that is a “fan” beam propagating in a direction indicated by Z axis **216**, wide in the vertical (Y) axis **214** and narrow in the horizontal (X) axis **212**, and second 1D linear array **108** forms receive main lobe **208** (also a “fan” beam) that is wide in the horizontal axis **212** and narrow in the vertical axis **214**. First 1D linear array **106** is controlled to steer transmit main lobe **206** (e.g., radar signal **180**) to any of a plurality of azimuth angles relative to horizontal axis **212** and second 1D linear array **108** is controlled to steer receiving of receive main lobe **208** (e.g., radar return signal **182**) to any of a plurality of elevation angles relative to vertical axis **214**. For example, radar controller **150** uses phase shifting (e.g., a variable signal delay for each element **202** of first 1D linear array **106**) to steer transmit main lobe **206** through a plurality of azimuth transmit angles defined relative to horizontal axis **212** and uses phase shifting (e.g., a variable signal delay for each element **204** of second 1D linear array **108**) to steer receive main lobe **208** through a plurality of elevation receive angles defined relative to vertical axis **214**. Radar controller **150** may independently steer transmit main lobe **206** and receive main lobe **208** through any number of angles and is not limited by the number of apertures (elements) **202/204**; however, the following examples illustrate the use of sixteen angular positions for each of transmit main lobe **206** and receive main lobe **208**. In one example of operation, radar controller **150** controls phase of each element **202** of first 1D linear array **106** to steer transmit main lobe **206** to any of sixteen different equally spaced elevation angles between a minimum elevation angle (e.g.,  $-45^\circ$ ) and a maximum elevation angle (e.g.,  $45^\circ$ ) and controls phase of each element **204** of second 1D linear array **108** to steer receive main lobe **208** to any of sixteen different equally spaced azimuth angles between a minimum azimuth angle (e.g.,  $-45^\circ$ ) and a maximum azimuth angle (e.g.,  $45^\circ$ ). Accordingly, through the pattern multiplication of transmit main lobe **206** and receive main lobe **208**, OLTRA radar system **100** achieves an effective pencil beam (e.g., spot beam **210**) that is directable within a two-dimensional (2D) array of angles (e.g., a  $16 \times 16$  angular array having two-hundred and fifty-six angular combinations). However, OLTRA antenna **104** may have more or fewer elements **202**, **204**, and may use more or fewer elevation and/or azimuth angles without departing from the scope hereof. Further, although the

number of transmit or receive angles is the same as the number of elements **202/204** in each of first 1D linear array **106** and second 1D linear array **108**, the number of transmit or receive angles may be different without departing from the scope hereof.

(21) FIG. **3A** shows an example 2D plot **300** of radar signal **180** of FIG. **1** illustrating a main transmit lobe **302** (e.g., transmit main lobe **206** of FIG. **2**) and a plurality of transmit sidelobes **304** (not all labelled for clarity of illustration) that generally diminish in intensity with increasing angle from main transmit lobe **302**. FIG. **3B** shows an example 2D plot **350** of receive beam characteristics (e.g., receive main lobe **208** of FIG. **2**) of second 1D linear array **108**, illustrating a main receive lobe **352** and a plurality of receive sidelobes **354** (not all labelled for clarity of illustration) that generally diminish in intensity with increasing angle from main receive lobe **352**. As appreciated, since first 1D linear array **106** is similar to second 1D linear array **108**, other than orientation, characteristics of main receive lobe **352** and a receive sidelobes **354** are similar to characteristics of main transmit lobe **302** and transmit sidelobes **304**, respectively.

(22) FIG. **4** is a 2D plot **400** illustrating a resultant pattern multiplication of main transmit lobe **302** and transmit sidelobes **304** of FIG. **3A** with main receive lobe **352** and receive sidelobes **354** of FIG. **3B**, illustrating spot beam **210** of FIG. **2** and resultant sidelobe products **404** (not all labelled for clarity of illustration). Sidelobe products **404** may cause clutter on radar display **192** unless blanked.

(23) Sidelobe Blanking for OLTRA

(24) In a conventional radar system (e.g., one that does not use OLTRA antenna **104**), a single omni-directional element, called the guard channel, is deployed with the main radar antenna. A signal level of a signal received by the guard channel is compared with a signal level of a return signal received by the main radar antenna to determine when the return signal includes echoes from sidelobes and should be “blanked.”

(25) One aspect of the present embodiments includes the realization that, since radar signal **180** (e.g., transmit main lobe **206**) is a fan beam, transmit sidelobes exist in only one dimension (e.g., repeated in the horizontal direction since first 1D linear array **106** is horizontal in the example of FIG. **2**), and therefore a conventional guard antenna (with relatively uniform antenna gain in all receive directions) can only disambiguate target returns from transmit sidelobes **304** (e.g., repeated in the horizontal dimension). That is, the conventional guard antenna cannot be used to blank receive sidelobes **354** of second 1D linear array **108** (e.g., returns caused by sidelobes in the vertical dimension) and therefore would not blank all clutter resulting from sidelobe products **404**. The present embodiments solve this problem by using both (a) a receive guard operation to blank signals within the receive sidelobes, and (b) a transmit guard operation to blank signals resulting from the transmit sidelobes. The receive blanking signal and the transmit blanking signal are used in combination to suppress clutter resulting from both the transmit sidelobes and the receive sidelobes (e.g., sidelobe products **404**), thereby improving an output of the OLTRA radar system **100**.

(26) To disambiguate target returns in the orthogonal dimension (e.g., horizontal axis **212**), OLTRA radar system **100** uses a single transmit “guard” element (e.g., transmit guard antenna **124**, FIG. **1**) to illuminate a scene (e.g., a space being evaluated by OLTRA radar system **100**) without sidelobes and uses the full receive linear array (e.g., second 1D linear array **108**, FIG. **2**, with sidelobes) to detect the return. In an alternative embodiment, OLTRA radar system **100** illuminates the scene without sidelobes by transmitting a “spoiled beam” using multiple elements of first 1D linear array **106** for increased effective isotropic radiated power (EIRP). Radar controller **150** controls second 1D linear array **108** to receive transmit guard signal **184** and receiver **140** sends first return guard signal **186** to signal processor **142** via transmission line **167**.

(27) Signal processor **142** (or receiver **140**) determines a first guard signal strength **146** of first return guard signal **186**, a second guard signal strength **147** of second return guard signal **188**, and a main beam signal strength **148** of radar return signal **182**. Blanker **144** compares (e.g., using a

thresholding algorithm) main beam signal strength **148** to first guard signal strength **146** and main beam signal strength **148** to second guard signal strength **147**. When blanker **144** determines that main beam signal strength **148** is either (a) not greater than first guard signal strength **146** or (b) not greater than second guard signal strength **147**, blanker **144** blanks radar return signal **182**. That is, a signal strength of radar return signal **182** is greater than first guard signal strength **146** and greater than second guard signal strength **147** when a reflection from a target in the scene is received through the main beam.

(28) FIGS. 5A and 5B show example perceived targets resulting from processing of radar return signal **182** without sidelobe blanking for a single corner reflector target positioned within a target space of OLTRA radar system **100**. FIG. 5A shows a 2D image graph **500** representing a plan view of perceived targets **502** (not all labelled for clarity of illustration) at 0° elevation over a distance of approximately ten meters from OLTRA antenna **104**, which is located at position 0,0 on the graph. Accordingly, 2D image graph **500** represents a horizontal slice through the target space of OLTRA radar system **100** as perceived by OLTRA radar system **100** without sidelobe blanking. Target **504** of 2D image graph **500** represents the single corner reflector target (e.g., the actual target within the target space of OLTRA radar system **100**), but is difficult to discern because of the clutter (false returns) in 2D image graph **500**.

(29) FIG. 5B is a range map plot **550** showing a 2D array of pixels for different azimuth and elevation angles relative to OLTRA antenna **104**, where each pixel represents a strongest radar return signal **182** above a threshold for any distance without sidelobe blanking. Those skilled in the art may represent both range and signal strength above a threshold to provide a better representation of the scene.

(30) Range map plot **550** indicates that radar return signal **182** includes many returns **552** (not all labelled for clarity of illustration) of varying intensity and direction and also includes stronger returns **554** at a corresponding location of the single corner reflector target within the target space of OLTRA radar system **100**.

(31) Clearly, as shown by 2D image graph **500** and range map plot **550**, without sidelobe blanking, false returns are perceived as targets and reduce quality of output from OLTRA radar system **100**.

(32) FIG. 6 is a graph **600** illustrating one example fan beam **602** that may represent transmit main lobe **206** and receive main lobe **208** of FIG. 2. Graph **600** shows the narrow beamwidth of fan beam **602** with a main lobe **608** (e.g., main transmit lobe **302** or main receive lobe **352**), and a plurality of sidelobes **610** (e.g., transmit sidelobes **304** or receive sidelobes **354**).

(33) Graph **600** also shows a transmit guard beam **604** that in this example is generated as a spoiled beam using multiple elements of first 1D linear array **106**. However, as shown in FIG. 1, transmit guard beam **604** may be generated by transmit guard antenna **124** (e.g., a single higher-powered element) that may be distinct from first 1D linear array **106**. Graph **600** also shows a receive guard beam **606** that may represent a receive beam (e.g., receive gain profile characteristics) of receive guard antenna **174** of FIG. 1. Graph **600** indicates that strength of the transmit guard signal **604** and receive guard signal **606** are higher than sidelobes **610** of fan beam **602** and thereby may be used to differentiate between sidelobe signals and main beam signals.

(34) Time Separated Operation

(35) FIG. 7 is a flowchart illustrating one example method **700** for OLTRA radar sidelobe blanking. Method **700** is implemented in blanker **144** of FIG. 1, for example. In this embodiment, a plurality of transmit antenna elements (e.g., apertures **202**) form a transmit array (e.g., first 1D linear array **106**) that extends linearly along a first axis (e.g., horizontal axis **212**), a plurality of receive antenna elements (e.g., apertures **204**) form a receive array (e.g., second 1D linear array **108**) that extends linearly along a second axis (e.g., vertical axis **214**) perpendicular to the first axis, and radar signal **180** and transmit guard signal **184** may have similar signal characteristic and are time separated during operation.

(36) In block **702**, method **700** transmits, with a transmit guard antenna having a wide beamwidth

in both axes, a transmit guard signal. In one example of block **702**, radar controller **150** activates guard transmitter **120** to output transmit guard signal **184** from transmit guard antenna **124**. In another example of block **702**, radar controller **150** controls multiple elements of first 1D linear array **106** to transmit a spoiled beam, similar to transmit guard signal **604** of FIG. **6**.

(37) In block **704**, method **700** receives a first return guard signal using the receive array with the main receive lobe at one receive angle of a plurality of receive angles defined relative to the second axis. In one example of block **704**, radar controller **150** controls second 1D linear array **108** to steer receive main lobe **208** to one of the plurality of receive angles defined relative to vertical axis **214** and receives first return guard signal **186**.

(38) In block **706**, method **700** determines a first guard signal strength of the first return guard signal. In one example of block **706**, signal processor **142** determines first guard signal strength **146** of first return guard signal **186**. In block **708**, method **700** transmits a radar signal using the transmit array with the main transmit lobe at one of a plurality of transmit angles defined relative to the first axis. In one example of block **708**, radar controller **150** controls first 1D linear array **106** to steer transmit main lobe **206** to one of the plurality of transmit angles defined relative to horizontal axis **212** and transmits radar signal **180**.

(39) In block **710**, method **700** receives a second return guard signal with a receive guard antenna having a wide beamwidth in both axes. In one example of block **710**, guard receiver **170** uses receive guard antenna **174** to receive second return guard signal **188**. In block **712**, method **700** determines a second guard signal strength of the second received signal. In one example of block **712**, signal processor **142** determines second guard signal strength **147** of second return guard signal **188**.

(40) In block **714**, method **700** receives a radar return signal using the receive array with the main receive lobe at the one receive angle. In one example of block **714**, radar controller **150** controls second 1D array **108** to steer receive main lobe **208** to the same receive angle as used in block **704** and receives radar return signal **182**. In block **716**, method **700** determines a main beam signal strength of the radar return signal. In one example of block **716**, signal processor **142** determines main beam signal strength **148** of radar return signal **182**.

(41) In block **718**, method **700** blanks the radar return signal when the main beam signal strength is less than either the first guard signal strength or the second guard signal strength. In one example of block **718**, blanker **144** blanks radar return signal **182** when main beam signal strength **148** is (a) less than first guard signal strength **146**, or (b) less than second guard signal strength **147**.

(42) Overlap Operation

(43) In this embodiment, radar signal **180** and transmit guard signal **184** are distinguishable from one another through differing signal characteristics (e.g., one or more of frequency, modulation, polarization, and encoding), and therefore need not be time separated during operation. That is, transmission of radar signal **180** and transmit guard signal **184** may occur concurrently (e.g., temporally overlapped).

(44) FIG. **8** is a flowchart illustrating one example method **800** for OLTRA radar sidelobe blanking with overlapping transmit signals. Method **800** is implemented in blanker **144** of FIG. **1**, for example.

(45) In block **802**, method **800** transmits, with a transmit guard antenna having a wide beamwidth in both axes, a transmit guard signal having first transmit signal characteristics. In one example of block **802**, radar controller **150** activates guard transmitter **120** to output transmit guard signal **184**, characterized by first characteristics, from transmit guard antenna **124**. In another example of block **702**, in certain embodiments, radar controller **150** controls multiple elements of first 1D linear array **106** to transmit a spoiled beam, characterized by first characteristics, similar to transmit guard beam **604** of FIG. **6**.

(46) In block **804**, method **800** transmits a radar signal having second transmit signal characteristics, different from the first transmit characteristics, using the transmit array with the

main transmit lobe at one of a plurality of transmit angles defined relative to the first axis. In one example of block **804**, radar controller **150** controls first 1D linear array **106** to steer transmit main lobe **206** to one of the plurality of transmit angles defined relative to horizontal axis **212** and transmits radar signal **180** that has second characteristics (e.g., one or more of frequency, modulation, polarization, and encoding) different from the first characteristics.

(47) In block **806**, method **800** receives a first received signal using the receive array having the main receive lobe at one receive angle of a plurality of receive angles defined relative to the second axis. In one example of block **806**, radar controller **150** controls second 1D linear array **108** to steer receive main lobe **208** to a first of the plurality of receive angles defined relative to vertical axis **214** and captures radar return signal **182** and first return guard signal **186**. In block **808**, method **800** determines a first guard signal strength of a portion of the first received signal having the first transmit characteristics. In one example of block **808**, signal processor **142** filters out radar return signal **182** and determines second guard signal strength **147** as a signal strength of first return guard signal **186**.

(48) In block **810**, method **800** receives a second received signal using a receive guard antenna having a wide beamwidth in both axes. In one example of block **810**, guard receiver **170** uses receive guard antenna **174** to receive second return guard signal **188**, which in this embodiment may include reflections of the radar signal **180** from an object or target (including volumetric targets, like weather) and reflections of transmit guard signal **184** from an object or target. In block **812**, method **800** determines a second guard signal strength of a portion of the second received signal having the second transmit characteristics. In one example of block **812**, signal processor **142** filters out first return guard signal **186** (e.g., reflections of transmit guard signal **184** from any objects or targets) and determines first guard signal strength **146** as a signal strength of reflections of radar signal **180**.

(49) In block **814**, method **800** extracts, from the second received signal, a radar return signal having the second transmit signal characteristics. In one example of block **814**, signal processor **142** filters out first return guard signal **186** (e.g., reflections of transmit guard signal **184** from any objects or targets) to determine radar return signal **182**.

(50) In block **816**, method **800** blanks the radar return signal when a signal strength of the radar return signal is (a) less than the first guard signal strength, or (b) less than the second guard signal strength. In one example of block **816**, blanker **144** blanks radar return signal **182** when main beam signal strength **148** is (a) less than first guard signal strength **146**, or (b) less than second guard signal strength **147**.

(51) FIGS. **9A** and **9B** show example perceived targets resulting from processing of radar return signal **182** with sidelobe blanking including a single corner reflector target positioned within the target space of OLTRA radar system **100**. FIG. **9A** shows an example 2D image graph **900** resulting from processing of radar return signal **182** with sidelobe blanking. 2D image graph **900** is a plan view showing a few perceived targets **902** at 0° elevation over a distance of approximately ten meters from OLTRA antenna **104**, which is located at the origin on the graph. Thus, 2D image graph **900** represents a horizontal slice through the target space of OLTRA radar system **100** as perceived by OLTRA radar system **100** when using sidelobe blanking. Target **904** of 2D image graph **900** represents the single corner reflector target.

(52) FIG. **9B** is one example range map plot **950** resulting from processing of radar return signal **182** with sidelobe blanking. Range map plot **950** is a 2D grid of pixels at different azimuth and elevation angles, of the strongest radar return signal **182** for any distance. Those skilled in the art may represent both range and signal strength above a threshold to provide a better representation of the scene. Range map plot **950** shows a few pixels **952** that indicate perceived targets, and pixels **954** represent the single corner reflector target. Clearly, with sidelobe blanking as shown by 2D image graph **900** and range map plot **950**, as compared to 2D image graph **500** and range map plot **550** of FIGS. **5A** and **5B**, the number of false returns perceived as targets is significantly reduced

by sidelobe blanking thereby improving the quality of output from OLTRA radar system **100**. (53) FIG. **10** is a schematic diagram illustrating an example computing apparatus **1000** that implements signal processor **142** of FIG. **1**, in embodiments. Computing apparatus **1000** includes a system bus **1001** that interconnects a digital processor **1004** with an input interface **1002**, an output interface **1006**, and memory **1008** (e.g., one or more of RAM, ROM, SRAM, DRAM, FLASH, etc.). Memory **1008** includes machine-readable instructions **1010** (e.g., software) that are executable by digital processor **1004** and data **1012**. In this embodiment, blanker **144** is implemented within machine-readable instructions **1010**; however, in other embodiments blanker **144** may be implemented as a combination of hardware and software. Data **1012** includes a digital representation of radar return signal **182**, first return guard signal **186**, second return guard signal **188**, first guard signal strength **146**, second guard signal strength **147**, and main beam signal strength **148**. (54) Although computing apparatus **1000** is shown with one system bus **1001**, computing apparatus **1000** may be implemented with a different type of architecture without departing from the scope hereof. For example, the machine-readable instructions **1010** and data **1012** may be stored in separate memories that communicate with digital processor **1004** using separate buses. In this case, the machine-readable instructions **1010** and data **1012** may be stored in separate memory spaces, thereby implementing a Harvard architecture. Alternatively, digital processor **1004** may include one or more layers of cache so that the computing apparatus **1000** implements a modified Harvard architecture using only system bus **1001**. In some embodiments, the machine-readable instructions **1010** are stored as an application in secondary storage (e.g., a hard drive), and loaded into memory **1008** upon powering on (i.e., boot up). In this case, machine-readable instructions **1010** and data **1012** share the same memory space, thereby implementing a von Neumann architecture. (55) Changes may be made in the above methods and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description or shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover all generic and specific features described herein, as well as all statements of the scope of the present method and system, which, as a matter of language, might be said to fall therebetween.

## Claims

1. A sidelobe blanking method for a radar system, the radar system having: a plurality of transmit antenna elements forming a transmit array that extends linearly along a first axis; a plurality of receive antenna elements forming a receive array that extends linearly along a second axis perpendicular to the first axis; the transmit array having a first radiation pattern with a main transmit lobe narrow along the first axis and wide along the second axis; and the receive array having a second radiation pattern with a main receive lobe wide in the first axis and narrow in the second axis; the sidelobe blanking method comprising: transmitting a guard signal from a transmit guard antenna having a wide beamwidth in both the first and second axes; controlling the receive array to sweep the main receive lobe through a plurality of receive angles defined relative to the second axis, wherein for each receive angle of the plurality of receive angles, the sidelobe blanking method includes: receiving a first received signal using the receive array; and determining a first guard signal strength of the first received signal for said each receive angle; and rastering the radar system by (i) controlling the transmit array to sweep the main transmit lobe through a plurality of transmit angles defined relative to the first axis and (ii) controlling the receive array to sweep the main receive lobe through the plurality of receive angles, wherein for each transmit angle of the plurality of transmit angles and each receive angle of the plurality of receive angles, the sidelobe blanking method includes: transmitting, with the transmit array, a radar signal; receiving, after said transmitting the radar signal, a second received signal with a receive guard antenna having a wide

beamwidth in both the first and second axes; determining a second guard signal strength of the second received signal; receiving, after said transmitting the radar signal, a third received signal with the receive array; determining a main beam signal strength of the third received signal; and blanking the radar signal when the main beam signal strength is less than either the first guard signal strength for said each receive angle or the second guard signal strength.

2. A sidelobe blanking method for a radar system, the radar system having: a plurality of transmit antenna elements forming a transmit array that extends linearly along a first axis; a plurality of receive antenna elements forming a receive array that extends linearly along a second axis perpendicular to the first axis; the transmit array having a radiation pattern with a main transmit lobe narrow along the first axis and wide along the second axis; and the receive array having a radiation pattern with a main receive lobe wide in the first axis and narrow in the second axis; and the sidelobe blanking method comprising: transmitting, with a transmit guard antenna having a wide beamwidth in both the first and second axes, a transmit guard signal; receiving a first return guard signal using the receive array with the main receive lobe at one receive angle of a plurality of receive angles defined relative to the second axis; determining a first guard signal strength of the first return guard signal; transmitting a radar signal using the transmit array with the main transmit lobe at one of a plurality of transmit angles defined relative to the first axis; receiving, after said transmitting the radar signal, a second return guard signal with a receive guard antenna having a wide beamwidth in both axes; determining a second guard signal strength of the second return guard signal; receiving, after said transmitting the radar signal, a radar return signal using the receive array with the main receive lobe at the one receive angle; determining a main beam signal strength of the radar return signal; and blanking the radar return signal when the main beam signal strength is less than either the first guard signal strength or the second guard signal strength.

3. The sidelobe blanking method of claim 2, further comprising rastering the radar system by: controlling the transmit array to sweep the main transmit lobe through a plurality of transmit angles defined relative to the first axis; and controlling the receive array to sweep the main receive lobe through the plurality of receive angles.

4. The sidelobe blanking method of claim 2, the radar system being a frequency-modulated continuous-wave radar system.

5. The sidelobe blanking method of claim 2, the transmit guard antenna being one of the plurality of transmit antenna elements.

6. The sidelobe blanking method of claim 2, the transmit guard antenna being at least two of the plurality of transmit antenna elements, the sidelobe blanking method further comprising transmitting the transmit guard signal as a spoiled beam.

7. The sidelobe blanking method of claim 2, the transmit guard antenna being distinct from the transmit array.

8. The sidelobe blanking method of claim 2, the receive guard antenna being one of the plurality of receive antenna elements.

9. The sidelobe blanking method of claim 2, the receive guard antenna being distinct from the receive array.

10. A sidelobe blanking method for a radar system having a plurality of transmit antenna elements forming a transmit array that extends linearly along a first axis and a plurality of receive antenna elements forming a receive array that extends linearly along a second axis perpendicular to the first axis, the transmit array having a radiation pattern with a main transmit lobe narrow along the first axis and wide along the second axis, the receive array having a radiation pattern with a main receive lobe wide along the first axis and narrow along the second axis, the sidelobe blanking method comprising: transmitting, with a transmit guard antenna having a wide beamwidth in both axes, a transmit guard signal having first transmit signal characteristics; transmitting a radar signal having second transmit signal characteristics, different from the first transmit signal characteristics, using the transmit array with the main transmit lobe at one of a plurality of transmit angles defined

relative to the first axis; receiving a first received signal with the receive array having the main receive lobe at one receive angle of a plurality of receive angles defined relative to the second axis; determining a first guard signal strength of a portion of the first received signal having the first transmit signal characteristics; receiving a second received signal with a receive guard antenna having a wide beamwidth in both axes; determining a second guard signal strength of a portion of the second received signal having the second transmit signal characteristics; extracting, from the first received signal, a radar return signal having the second transmit signal characteristics; and blanking the radar return signal when a signal strength of the radar return signal is either (a) less than the first guard signal strength, or (b) less than the second guard signal strength.

11. The sidelobe blanking method of claim 10, further comprising rastering the radar system by: controlling the transmit array to sweep the main transmit lobe through a plurality of transmit angles defined relative to the first axis; and controlling the receive array to sweep the main receive lobe through the plurality of receive angles.

12. The sidelobe blanking method of claim 10, the radar system being a frequency-modulated continuous-wave radar.

13. The sidelobe blanking method of claim 10, the transmit guard antenna being one of the plurality of transmit antenna elements.

14. The sidelobe blanking method of claim 10, the transmit guard antenna being multiple transmit antenna elements of the plurality of transmit antenna elements, the method further comprising transmitting the transmit guard signal as a spoiled beam from the multiple transmit antenna elements.

15. The sidelobe blanking method of claim 10, the transmit guard antenna being distinct from the transmit array.

16. The sidelobe blanking method of claim 10, the receive guard antenna being one of the plurality of receive antenna elements.

17. The sidelobe blanking method of claim 10, the receive guard antenna being distinct from the receive array.

18. The sidelobe blanking method of claim 10, the second received signal including a reflection of the transmit guard signal and a reflection of the radar signal.

19. The sidelobe blanking method of claim 10, the first transmit signal characteristics and the second transmit signal characteristics differing in one or more of frequency, modulation, polarization, and encoding.

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